

Tom Lin

HEDERA ZK VOTING

Project Overview

- Privacy-preserving, tamper-proof, and verifiable voting system
- Built on Hedera using zero-knowledge proofs (ZKP)
- Public voter registry and private vote submissions published to two Hedera Consensus Service topics
- Efficient Groth16 circuit guarantees voter eligibility and one-vote-per-person

Tech Stack

- Frontend
 - Next.js 16
 - TypeScript
 - Tailwind CSS
 - Hedera Wallet Connect
- Backend
 - Hedera Consensus Service
 - Circom 2
 - snarkjs (Groth 16)
 - Merkle Trees

Demo

<https://youtu.be/NHrLTUVhxbk>

Future Roadmap

- Improve UI to support other wallet connectors like Blade, Metamask etc.
- Multi-poll support with dynamic poll creation, independent Merkle trees, and independent nullifiers
- Decentralized voter registry management using threshold signatures or DAO-controlled enrollment